Off-Path Attacking the Web

Off-Path Attacking the Web - Author not stated, usenix.org.

- Published: date not stated
- Original: <https://www.usenix.org/conference/woot12/workshop-program/presentation/gilad>
- Preserved from: https://www.usenix.org/conference/woot12/workshop-program/presentation/gilad (live) on 2026-08-08
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Off-Path Attacking the Web

Yossi Gilad and Amir Herzberg, *Bar Ilan University* ***Awarded Best Student Paper!***

We show how an off-path (spoofing-only) attacker can perform cross-site scripting (XSS), cross-site request forgery (CSRF) and site spoofing/defacement attacks, without requiring vulnerabilities in either web-browser or server, and circumventing known defenses. The attacks are practical and require a puppet (malicious script in browser sandbox) running on a victim client machine, and an attacker capable of IP-spoofing on the Internet.

Our attacks are based on a technique that allows an offpath attacker to efficiently learn the sequence numbers of both the client and server in a TCP connection. This technique exploits the fact that many computers, in particular those running (any recent version of) Windows, use a global IP-ID counter, which provides a side channel allowing efficient exposure of the connection sequence numbers.

We present results of experiments evaluating the learning technique and the attacks that exploit it. We also present practical defenses that can be deployed at the firewall level, either at the client or server end; no changes to existing TCP/IP stacks are required.

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BibTeX

@inproceedings {179508, title = {{Off-Path} Attacking the Web}, booktitle = {6th USENIX Workshop on Offensive Technologies (WOOT 12)}, year = {2012}, address = {Bellevue, WA}, url = {https://www.usenix.org/conference/woot12/workshop-program/presentation/Gilad}, publisher = {USENIX Association}, month = aug }

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